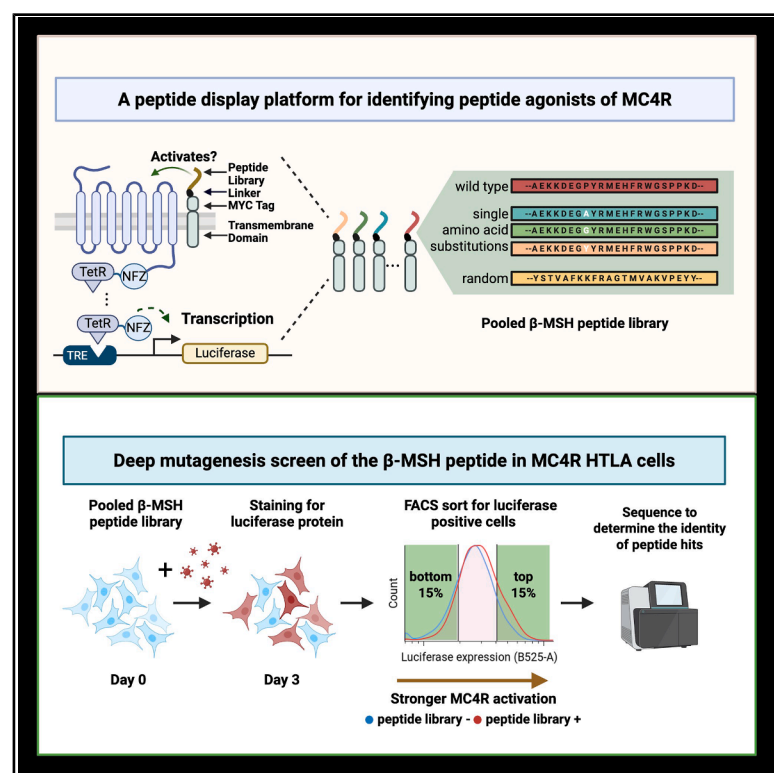


A peptide display system identifies a potent mutant β -melanocyte-stimulating hormone agonist of melanocortin-4 receptor

Graphical abstract



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In brief

Lin et al. developed a high-throughput cell surface peptide display platform and coupled it with a β -arrestin-based melanocortin-4 receptor reporter to rapidly screen over 2,000 β -melanocyte-stimulating hormone variants. This approach uncovers key peptide mutations influencing receptor activation, including a novel D5H mutant with enhanced potency.

Highlights

- High-throughput peptide display platform for screening MC4R ligands
- β -MSH deep mutagenesis reveals key residues for MC4R interaction
- Identification of residues important for the processing of β -MSH
- Discovery of a β -MSH D5H variant with enhanced MC4R activation



Short article

A peptide display system identifies a potent mutant β -melanocyte-stimulating hormone agonist of melanocortin-4 receptor

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SUMMARY

Non-olfactory G-protein-coupled receptors (GPCRs) regulate vital physiological functions and are targets for ~34% of US Food and Drug Administration (FDA)-approved drugs. While small-molecule-activated GPCRs are well studied, there is growing interest in peptide GPCRs, particularly the melanocortin-4 receptor (MC4R), a key regulator of energy balance and appetite. Activation of MC4R by β -melanocyte-stimulating hormone (β -MSH) reduces food intake, and pathway dysfunction leads to obesity. However, current methods to study GPCR-peptide interactions are resource intensive and low throughput. To address this, we developed a high-throughput cell surface peptide display platform with a β -arrestin-based MC4R reporter to screen over 2,000 β -MSH point mutants. This approach identified peptide variants that significantly impact MC4R activation, including a novel D5H mutant with enhanced receptor activation. Our results demonstrate a scalable method to directly link GPCR activation to peptide variants, offering insights for therapeutic peptide design.

INTRODUCTION

G-protein-coupled receptors (GPCRs) are the largest class of membrane proteins in the human genome, composed of over 800 receptors. 398 of these receptors are non-olfactory GPCRs integral in regulating numerous biological functions, including sensory perception, immune response, and neurotransmission.¹ Given their extensive involvement in critical physiological processes, non-olfactory GPCRs are key targets for drug development, accounting for over 34% of all US Food and Drug Administration (FDA)-approved drugs.² Most of the GPCRs targeted by FDA-approved drugs are aminergic with small-molecule ligands.^{1,2} However, in an interesting shift in focus, more than half of the emerging GPCR targets that have reached clinical trials but have not yet been approved are peptide receptors.^{1,2} The expansion in the number of peptide receptors in clinical trials has likely been enabled by exciting technological advances. These include more comprehensive annotation of natural peptide ligands and emerging novel strategies such as half-life extension platforms and enhancing resistance to proteolysis, which have improved pharmacokinetics and oral bioavailability.^{2,3}

The melanocortin-4 receptor (MC4R) is one of five human melanocortin receptors (MC1–5R) that belong to a subgroup of class A GPCRs.⁴ The melanocortin receptors were originally named for their earliest known function in melanogenesis and have since

been shown to play roles in energy homeostasis, cardiovascular function, immune regulation, and sexual functions.^{5–7} MC4R plays a central role in regulating energy balance and appetite.⁴ Given its role in regulating satiety, MC4R is one of the major peptide GPCRs being targeted by novel therapeutics designed to counteract obesity.^{8,9} Activation of MC4R in neurons of the paraventricular nucleus of the hypothalamus by its natural agonist, β -melanocyte-stimulating hormone (β -MSH), leads to reduced appetite and decreased food intake.⁴ Loss of function of MC4R in other species, such as rats, mice, and canines, increases appetite and weight gain, demonstrating the conservation of the MC4R- β -MSH axis in regulating energy homeostasis and food motivation.^{10–12} Furthermore, naturally occurring mutations in the MC4R and β -MSH pathway are the most frequent cause of monogenic obesity in humans, with a prevalence of 2.5%–8.3% in children with early-onset obesity.^{13–15} These findings underscore the critical role of the MC4R and β -MSH pathway in energy regulation and illustrate the clinical significance of characterizing the interaction of β -MSH with MC4R.

Several approaches exist to study GPCR-peptide interactions; however, existing methods have some key limitations that limit their utility in the systematic study of GPCR-peptide interactions. Phage display, a widely used assay for identifying proteins that bind to specific receptors, has been critical in the development of biological drugs.^{16–19} In this process, peptide libraries are displayed on the surface of phages and screened against a



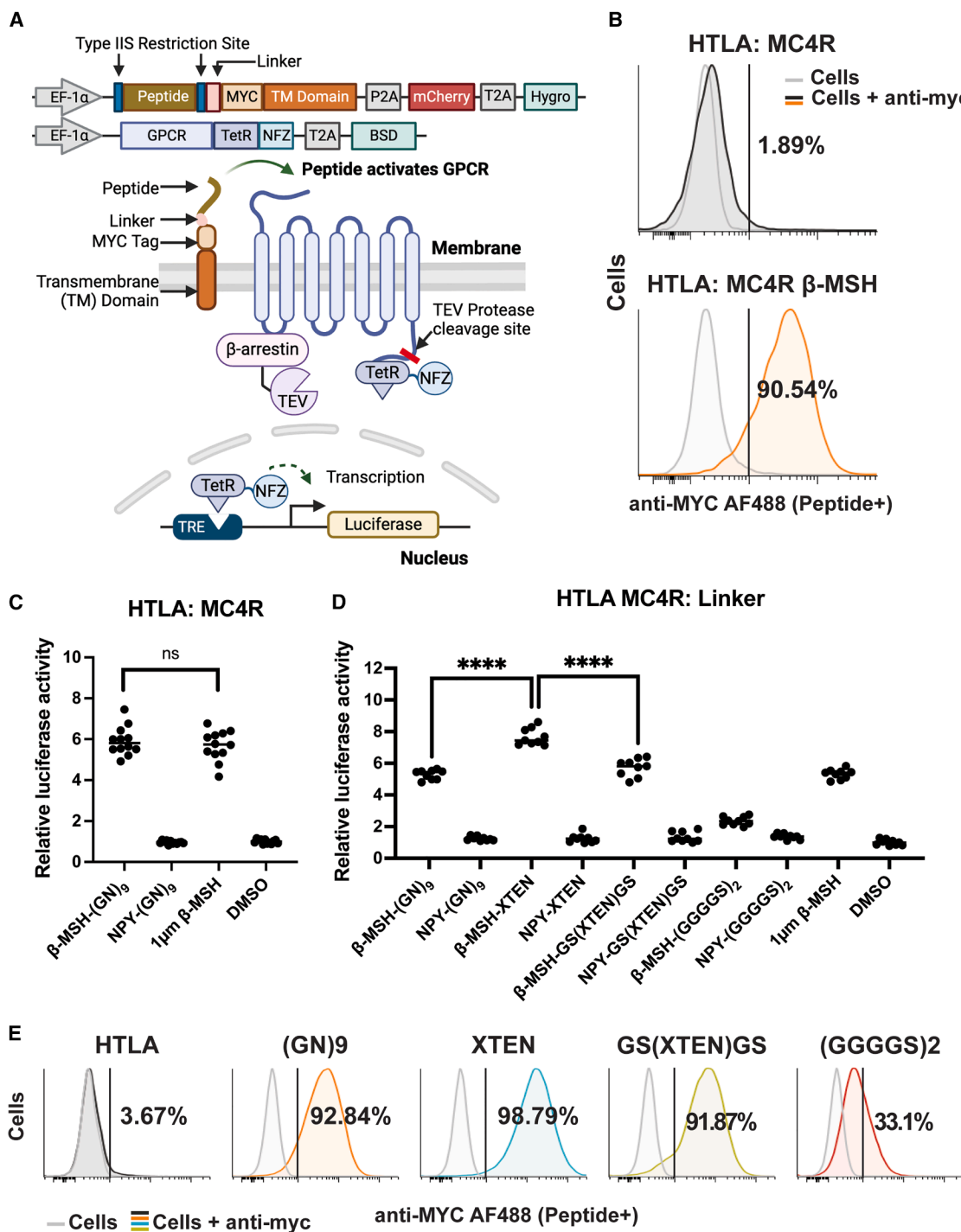


Figure 1. Development of a peptide display system to identify agonists of G-protein-coupled receptors

(A) Schematic illustrating a cell-based platform to identify peptide agonists against human G-protein-coupled receptors (GPCRs). The system consisted of (1) HTLA reporter cells that expressed β-arrestin fused to tobacco etch virus (TEV) protease and a tetracycline responsive element (TRE) driving luciferase expression, (2) a synthetic GPCR with a TEV protease cleavage site linked to a Tet repressor (TetR)-NFZ tripartite activator, and (3) a peptide expression vector with a type IIS restriction site fused to a flexible linker, MYC tag, and platelet-derived growth factor receptor β (PDGFRβ) transmembrane (TM) domain. Proper co-expression of a GPCR with its cognate peptide agonist triggers the cleavage of the TetR-NFZ activator and turns on the luciferase reporter. (B) Cell surface measurement of β-MSH peptide expression in MC4R HTLA cells using an anti-myc tag conjugated antibody. A glycine and asparagine (GN)₉ repeat linker was used. Untransfected MC4R HTLA cells are stained in parallel as a negative control.

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target protein. After screening, the binding clones are isolated and sequenced to identify putative peptide hits.²⁰ However, because phage display is a binding-based assay, further steps such as testing the effect of synthesized candidate peptides in functional assays are necessary. The workload to identify functional binders from this pool of physical binders is significant, and many clones selected as strong binders from natural peptide libraries are nonfunctional.²¹ To address this, cell-based functional assays have been developed in which cells express a reporter system to indicate the activity of GPCRs of interest.²² Panels of peptides can be tested in an arrayed manner on the reporter cell lines, leading to the identification of novel agonists of GPCRs.^{22,23} However, these assays are limited by the scale and cost of plate-based robotics and assays, as well as the high cost of peptide synthesis. It is worth noting that novel approaches, such as combinatorial chemical synthesis methods and RNA display, have been developed to address the cost and complexity of peptide library synthesis, particularly for macrocyclic peptides due to their favorable pharmacokinetic properties.^{24,25} These methods have been instrumental in identifying cyclic peptides that target therapeutic pathways, including those implicated in hepatic fibrosis and thrombin regulation.^{25,26} However, their potential application in screening linear peptides against GPCRs remains largely unexplored. Lastly, while genetic association studies have advanced our understanding of the clinical relevance of peptide and receptor variants,^{10,27,28} they are limited in their ability to identify rare variants that might not be represented in the sampled populations.

Here, we report the development of a peptide display platform that enables high-throughput identification of peptide ligands that agonize a specific GPCR. This method integrates the pooled synthesis of oligonucleotide libraries encoding variants of the peptides of interest, a peptide display system that expresses and localizes these peptides on the cell surface, and a synthetic β -arrestin-based cell recruitment reporter for readout. We validated this system using MC4R and its known ligand, β -MSH. We then created a library to perform deep mutational scanning of the β -MSH peptide, simultaneously screening over 2,000 peptide elements in a single pooled experiment. We identified novel single-amino-acid mutations in β -MSH that significantly enhance or diminish the ability of β -MSH to activate MC4R. Linking the mutant β -MSH peptides to publicly available clinical data, we discovered that one of the identified loss-of-function mutants, a glutamic acid [E] to glycine [G] mutation in the second position of β -MSH, is prevalent in children with obesity.²⁹ Additionally, we identified a novel D5H mutation in β -MSH that enhances its ability to activate MC4R. Together, these results demonstrate a single-cell approach that, for the first time, allows

the simultaneous measurement of activities for thousands of peptide variants to identify potent agonists of GPCRs.

RESULTS

Design of the peptide display platform to identify GPCR agonist

To develop a high-throughput platform to identify peptide agonists of GPCRs, we first sought to (1) devise a strategy that would allow the expression of a library of peptides on the cell membrane and (2) develop a cell-based reporter that would respond when a peptide ligand agonizes the GPCR of interest. To express a library of peptides that can interact with GPCRs on the cell surface, we created a modular peptide display system that enables the expression and localization of any peptide of interest to the cell membrane. At the 5' end of the peptide display construct, we incorporated a cleavable signal peptide derived from pre-pro-trypsin³⁰ to promote membrane localization and a peptide cloning site for efficient cloning. At the 3' end, we added a platelet-derived growth factor receptor β (PDGFR β) transmembrane domain to embed the construct into the cell membrane and a MYC epitope tag to facilitate monitoring of cell surface expression by flow cytometry. The components are connected by a linker to ensure proper domain folding (Figure 1A).

To ensure the scalability and generalizability of our platform across a broad range of GPCRs, we paired our peptide display system with a modified version of the widely used PRESTO-Tango GPCR reporter, an assay that measures β -arrestin recruitment activity that has been validated in 125 nonorphan GPCRs.²² While GPCRs can signal through pathways other than β -arrestin recruitment, it has been shown that 88% of the variance in the association between MC4R variants and clinical phenotypes, such as BMI, is attributed to their effect on β -arrestin recruitment. Additionally, individuals carrying gain-of-function MC4R alleles with signaling bias tend to exhibit increased β -arrestin recruitment relative to other signaling pathways.^{10,31} These findings support our rationale for using a β -arrestin-based reporter to measure MC4R activation.

Briefly, in our system, each GPCR is modified to include a Tango protein fragment at the 3' end, consisting of a tobacco etch virus (TEV) protease cleavage site and a tetracycline-controlled transactivator (tTA). This synthetic GPCR is transfected into HTLA cells, a HEK293-derived reporter cell line stably expressing a luciferase reporter and a synthetic β -arrestin fused to the TEV protease. Upon activation by a ligand, the GPCR triggers the recruitment of a synthetic β -arrestin fused to the TEV protease and releases the tTA.³² Once the tTA is freed, it translocates into the nucleus and turns on the reporter, leading to

(C) Luciferase reporter assay measuring the relative activation of the MC4R in HTLA cells co-expressing synthetic MC4R and its cognate peptide, β -MSH. MC4R HTLA cells transduced with NPY are the negative control, while MC4R HTLA cells treated with synthetic β -MSH are the positive control. All measurements were normalized to the mean of MC4R HTLA cells treated with DMSO.

(D) Relative luciferase reporter activity of HTLA cells co-expressing synthetic MC4R and β -MSH fused to various flexible linkers. MC4R HTLA cells transduced with NPY fused to the corresponding linkers are used as the negative control. MC4R HTLA cells treated with synthetic β -MSH are used as the positive control. All measurements were normalized to the mean luciferase activity of MC4R HTLA cells treated with DMSO. Statistical significance is indicated as follows: **** $p < 0.0001$.

(E) Measurement of β -MSH protein abundance in MC4R HTLA cells transduced with β -MSH tethered to various flexible linkers. Untransduced MC4R HTLA cells are used as a negative gating control.

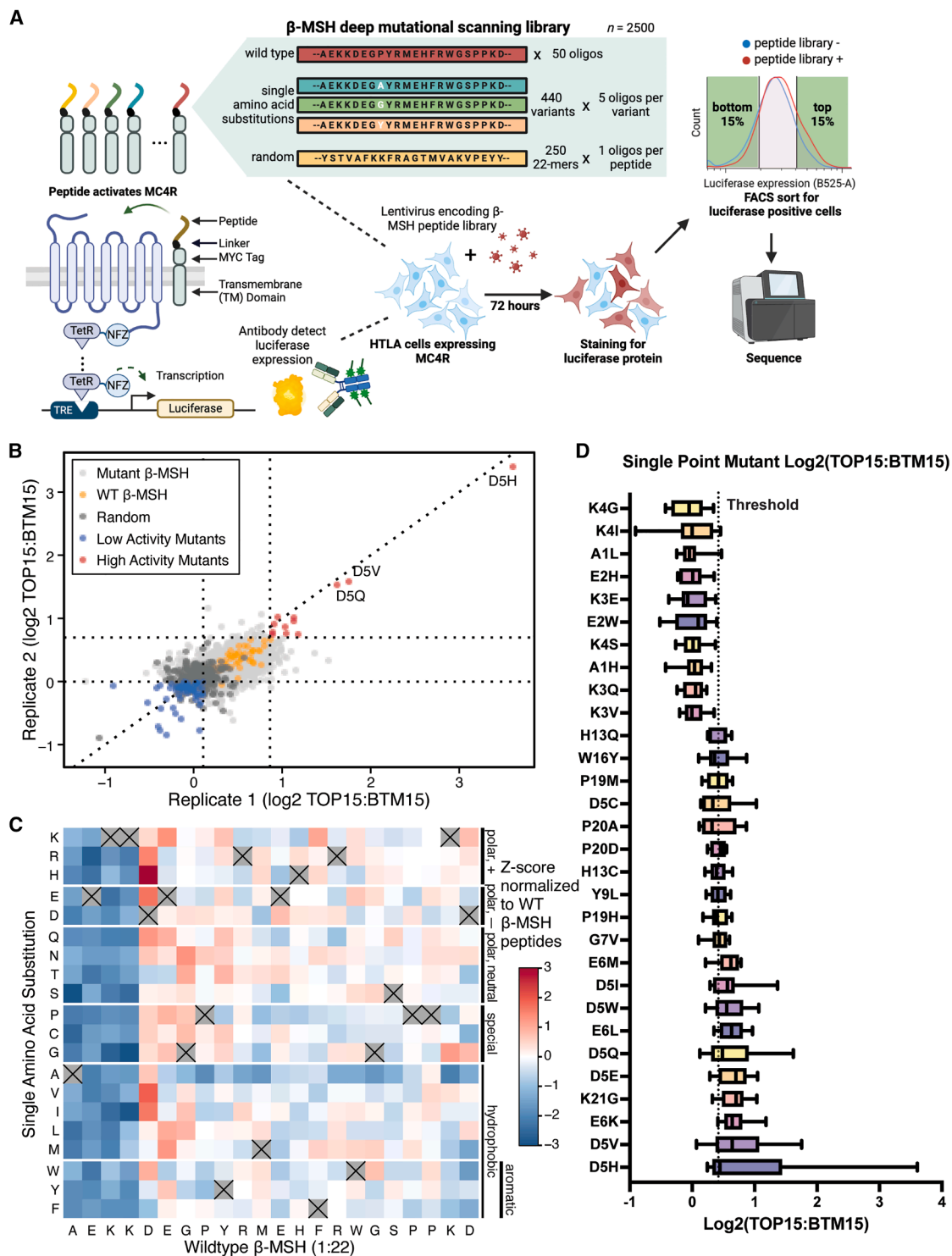


Figure 2. Deep mutational scan of the β -MSH peptide in MC4R HTLA cells using the peptide display system

(A) Schematic illustrating the composition of the β -MSH peptide library and the deep mutagenesis screening workflow.

(B) Reproducibility from two biological replicates. In green are wild-type (WT) β -MSH peptides, in dark gray are random peptides, in yellow are WT β -MSH peptide, in light gray are single-nucleotide mutants of β -MSH, in red are high-activity β -MSH mutants, and in blue are low-activity β -MSH mutants. Dashed lines represent two standard deviations from the mean of the $\log_2$ (TOP15:BTM15) values of the WT β -MSH peptides.

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luciferase expression (Figure 1A). It has been noted in the literature that Tet-based reporter systems experience leakiness and high expression of VP16, which can lead to cellular toxicity.^{33,34} Significant advances have been made to address these challenges, including the development of new activator domains and mutations in the Tet repressor (TetR) protein to reduce leakage.^{34,35} To improve the signal-to-noise ratio of the reporter, we modified the Tango component by replacing the original tTA with tTA Advance,³⁴ a tTA with improved stability and minimal background expression. Furthermore, we replaced the VP16 activator with a more compact human activator combination, NFZ, that we developed,³⁶ which has stronger activation and, when overexpressed, is less toxic to human cells (Figure 1A). This modified GPCR Tango construct was encoded within a lentiviral vector and transduced into HTLA cells to create stable reporter cell lines.

Validation and optimization of the GPCR peptide display platform

To assess the ability of the peptide display construct to localize peptides of interest to the cell membrane and to determine if co-expression of a GPCR and peptide ligand is sufficient to activate the reporter, we performed preliminary tests of our peptide display platform using a known peptide-GPCR pair: the MC4R and its natural peptide ligand, β -MSH.³⁷ We lentivirally transduced and selected for a pure population of MC4R stably expressing HTLA cells with peptide display constructs encoding β -MSH. We measured peptide expression at the cell surface through anti-Myc staining and found that more than 90.54% of the transduced cells stained positive for cell surface expression of β -MSH post-transduction (Figure 1B). After confirming the proper localization of the peptide to the membrane, we then assayed whether the peptide display of β -MSH is sufficient to activate MC4R and induce reporter expression of luciferase. We found that cells expressing peptide-displayed β -MSH were able to induce similar levels of luciferase activity as MC4R cells treated with 1 μ M of the synthetic form of the peptide ($****p < 0.0001$, Student's *t* test; Figure 1C; Data S1). Importantly, MC4R HTLA cells expressing a negative control peptide expected to have no activity on these cells, neuropeptide Y (NPY), did not induce significant luciferase expression.

To evaluate whether our peptide display system serves as an effective reporter system for measuring the activation of other GPCRs by their putative peptide ligands, we repeated the experiment with three additional GPCRs and their corresponding peptide agonists. We observed that the co-expression of certain peptides with their respective receptors induced a significant increase in luciferase expression, such as NPY with NPY receptor 5 (NPY5R) and motilin with the motilin receptor (MLNR). Conversely, we found that the peptide display of some peptides, such as RFamide-related peptide-1 (RFRP-1), did not induce significant luciferase expression when co-expressed with their

putative GPCR (in the case of RFRP-1, neuropeptide FF receptor 1 [NPFF1]) (Figure S1).

Next, we sought to understand how the linker sequence in the peptide display vector influences peptide activation of the GPCR, as protein linkers have previously been reported to influence protein folding and stability.³⁸ Our preliminary experiments were conducted using a (GN)9 linker, which we extended by testing three additional linkers (Table S1).^{38,39} We found that the XTEN linker increased the abundance of β -MSH present at the cell surface and led to a significant increase in MC4R activation compared to the other linkers (Figures 1D and 1E; Data S1). These results are consistent with previous observations that the XTEN linker can improve the stability of fusion proteins.^{39,40} However, the incorporation of amino acids that should increase linker flexibility, such as glycines and serines, upstream and downstream of the XTEN linker decreased peptide abundance at the cell surface and reduced reporter activation. The inclusion of hyper-flexible linkers composed of only glycine and serines decreased cell surface β -MSH abundance by more than 2-fold and luciferase reporter activation by more than 3-fold (Figures 1D and 1E; Data S1). Based on these results, we determined that XTEN was the optimal linker to use in the context of our peptide display system.

Designing a deep mutational scan of the β -MSH peptide

We next aimed to explore whether the peptide display platform could be expanded to perform high-throughput screening and identify peptide agonists of GPCRs in a pooled fashion. To test this, we designed an “all-by-all” deep mutational scanning library of the β -MSH peptide, testing every single possible point mutation across all 22 positions of the peptide. Each mutant was encoded using five different oligonucleotide sequences, varying codon usage to enable the unique determination of each sequence by deep sequencing.⁴¹ We also generated 50 oligos each encoding the wild-type β -MSH peptide and 250 22-mer random peptides.⁴² In total, the library was composed of 2,500 elements encoding 740 unique peptides (Table S2). We transduced the library into MC4R-expressing HTLA reporter cells (Figure 2A). Then, the cells were permeabilized, stained with an anti-luciferase antibody, and sorted into high (top 15% expression, TOP15) and low (bottom 15% expression, BTM15) luciferase expression-level bins. We sorted a minimum of 1 million cells into each bin to maintain a coverage of 400 $\times$ per element. We sequenced the two bins and calculated the enrichment of the peptides in each sorted population. Encouragingly, we observed an increase in luciferase levels in the MC4R-expressing HTLA cells that expressed the peptide library compared to the cells that did not. For example, in both biological replicates, we saw a 2-fold increase in luciferase-positive cells in the population expressing the peptide library compared to non-transduced cells (Figures 2C and S2).

After filtering out peptide oligos with low reads, we conservatively called peptides that fall two standard deviations above and

(C) Heatmap of the normalized Z scores of the average $\log_2$ (TOP15:BTM15) values of mutants at each position. The Z scores are normalized to the mean and standard deviation of the WT β -MSH $\log_2$ (TOP15:BTM15) values.

(D) Whisker plot of the $\log_2$ (TOP15:BTM15) values of all oligos encoding top ten high-, neutral-, and low-activity mutants. The dotted line indicates the threshold value that corresponds to the mean of the WT β -MSH peptide $\log_2$ (TOP15:BTM15) values. The whiskers represent the range of $\log_2$ (TOP15) values for each point mutation.

below the mean of the $\log_2(\text{TOP15:BTM15})$ score of all the wild-type β -MSH peptides high- and low-activity mutants, respectively (Figure 2B; Data S2). We found that over 74% of the random negative control peptides did not significantly activate MC4R, with their activity falling at least two standard deviations below the mean of the wild-type peptide in at least one biological replicate. Based on the enrichment scores of all the wild-type positive control peptides and random sequence negative control peptides, we determined that our screen had an excellent signal-to-noise ratio (Z prime value = 0.89 for both biological replicates). We compared the overlap of significant hits between the two biological replicates to assess the reproducibility of the screens. The results showed significant overlap in hits (Fisher's exact test $p = 9.52 \times 10^{-22}$), indicating good reproducibility. Overall, the consistent reproducibility, excellent signal-to-noise ratio, and accurate identification of expected controls across both biological replicates suggest that the GPCR peptide display platform yielded reliable results.

Validating novel β -MSH mutants

While we found that most point mutations do not significantly impact the ability of β -MSH to activate MC4R, we did observe a striking enrichment in loss-of-function mutations in the first four amino acids of the β -MSH peptide (Figures 2C and 2D; Data S2 and S3). In fact, all of the top-ten-ranking low-activity mutants identified in the screen bore mutations in the first four positions of the peptide (Figure 2D; Data S2). We further validated our findings by transducing four of the candidate loss-of-function mutants into MC4R HTLA cells in a low-throughput experiment and measured luciferase activity. We found that reporter cells expressing all of these mutants failed to activate MC4R and induce significant luciferase expression (Figure 3B; Data S4). Interestingly, the glutamic acid [E]-to-glycine [G] mutation in the second position of the peptide is more prevalent in children with obesity, though the mechanism by which this missense mutation causes obesity is still unidentified.²⁹ β -MSH is derived from a precursor polypeptide, pro-opiomelanocortin, through multiple cleavage events at dibasic amino acids, and mutations in those cleavage positions have been previously reported to contribute to obesity in children.^{43,44} Using signalP 6.0, we found two predicted cleavage sites within the first four amino acids for the wild-type β -MSH peptide.⁴⁵ We hypothesized that these four positions are important for peptide cleavage and conversion of the β -MSH peptide from the pro-form into the active form (Figure 3A). To test this hypothesis, we treated MC4R-expressing HTLA cells with wild-type β -MSH and a shortened version of β -MSH without the first four amino acids. There was no significant difference in MC4R reporter activation between the two peptide treatments (Figure 3C; Data S4). This suggests that the first four amino acids are not important for binding to MC4R and that the glutamic acid-to-glycine SNP mutation in the second position likely disrupts the proteolytic processing of wild-type β -MSH.

Additionally, mutations that significantly enhance β -MSH's ability to activate MC4R are rare (Figure 2C; Data S3). Nine out of the ten high-activity mutants bear mutations in the 5th and 6th positions of the β -MSH peptide (Figure 2D; Data S2). We transduced the D5H, D5V, and E6K mutants into MC4R-ex-

pressing HTLA cells and found that they indeed were able to activate MC4R more potently compared to the wild type. Notably, an aspartic acid [D]-to-histidine [H] substitution in the fifth position increased MC4R activation by more than 50% when expressed on a peptide display construct (Figure 3C; Data S4). To eliminate the possibility that the enhanced reporter activation is due to a unique property of the peptide being expressed on the display construct, we also treated reporter cells with a chemically synthesized version of the D5H mutant peptide. As expected, the D5H mutant peptide demonstrated increased MC4R activation compared to the wild-type β -MSH (Figure 3D; Data S4). The EC50 value of the D5H peptide was 1.46 nM, significantly lower than the EC50 value of the wild-type β -MSH, which was 8.23 nM. These findings illuminate the critical roles of specific amino acid residues in β -MSH in regulating MC4R function and identify candidate residues for engineering enhanced activity peptides.

To investigate the structural basis for the increased potency of the D5H variant in activating MC4R compared to the wild-type peptide, we performed receptor-peptide co-folding analysis using AlphaFold 3.⁴⁶ This analysis was conducted with MC4R and the cleaved D5H variant (predicted template modeling score [pTM] = 0.79; interface pTM [ipTM] = 0.77), as well as the cleaved wild-type peptide (pTM = 0.76; ipTM = 0.79), excluding the first four amino acid pro-sequences.⁴⁷ As a reference, we used the published cryo-electron microscopy (cryo-EM) structure of the MC4R bound to Nle4-D-Phe7- α -melanocyte-stimulating hormone (NDP- α -MSH), a peptide with high homology to β -MSH (Figure S3A).⁴⁸ The NDP- α -MSH and β -MSH peptides both have nanomolar EC50 values against MC4R and share a His-x-Arg-Trp [H-x-R-W] pharmacophore (Figure S3A).^{48,49} Overlaying the AlphaFold 3-predicted structures with the NDP- α -MSH-MC4R complex structure revealed that the predicted MC4R structures aligned closely with the published cryo-EM structure. AlphaFold 3 flagged the predicted structure of the N-terminal domain (amino acids 1–39) of MC4R as having very low confidence (predicted local distance difference test [pLDDT] < 50). The N terminus of MC4R has been found to be irrelevant to interactions with NDP- α -MSH.⁴⁸ Attempts to characterize the full-length MC4R structure using cryo-EM were unable to resolve this N-terminal domain, likely due to its unstructured nature. Furthermore, a recent deep mutagenesis study confirmed that mutations in the N terminus do not significantly disrupt MC4R signaling.⁵⁰ Consequently, we have hidden the N terminus in our analysis. The β -MSH peptides were predicted to interact with MC4R at the same binding site as NDP- α -MSH, with the [H-x-R-W] pharmacophore representing the deepest point of interaction for all three peptides (Figure S3B). However, no structural differences were observed between the wild-type β -MSH peptide and D5H mutant peptides. The amino acid substitution in the D5H variant occurs outside the binding interface with the MC4R, suggesting that the increased potency of the D5H variant may arise from mechanisms other than structural changes in peptide interactions with MC4R (Figure S3B). In the wild-type β -MSH peptide, aspartic acid [D] is a polar, negatively charged amino acid, while the histidine [H] substitution in the mutant peptide introduces a polar, positively charged amino acid. The C terminus of the wild-type peptide consists of lysine [K] and aspartic acid [D], which are polar, positively and

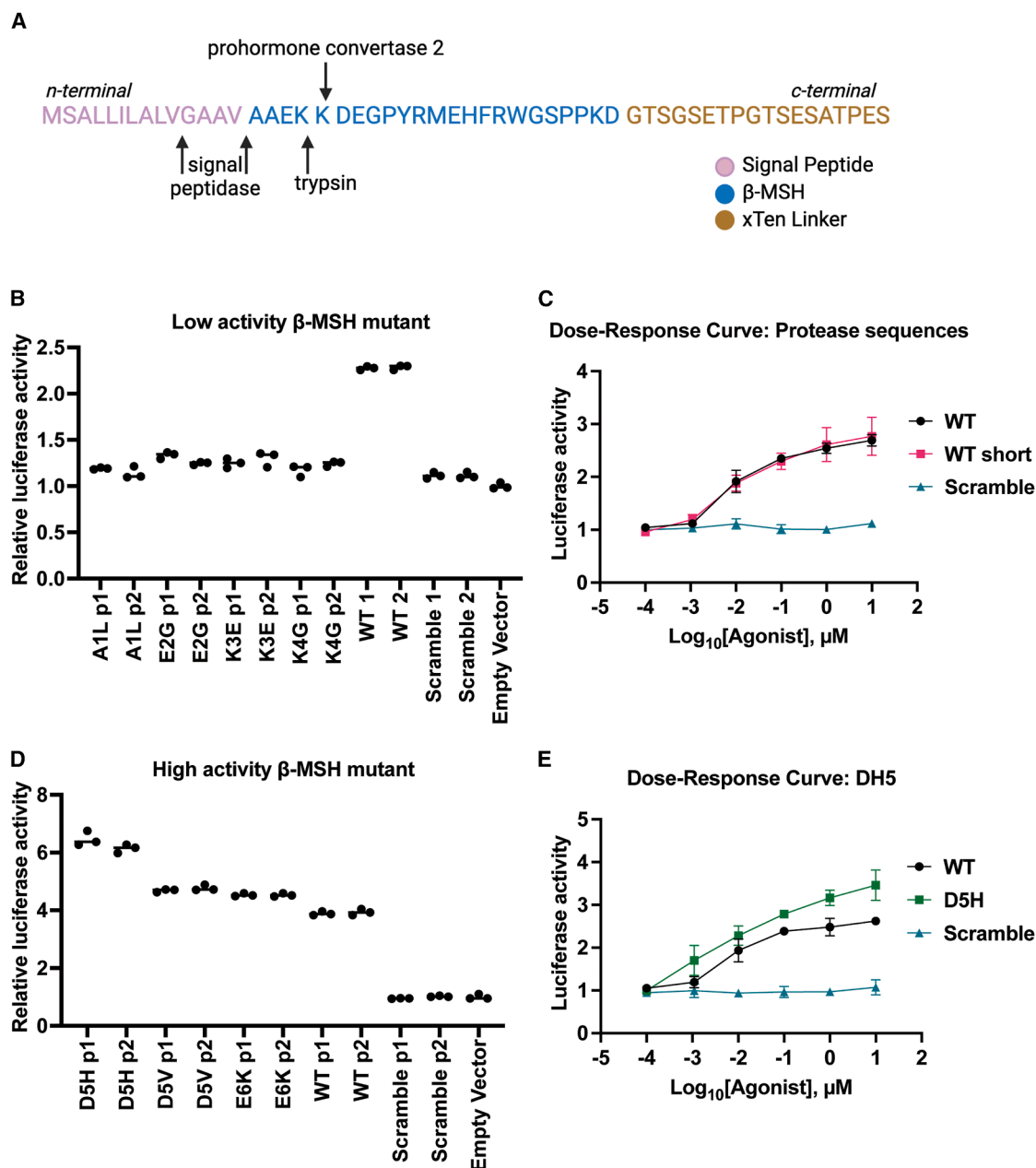


Figure 3. Identification and validation of a β-MSH mutant with enhanced ability to activate MC4R

(A) Diagram of the amino acid sequence of the signal peptide (red), β-MSH peptide (blue), and Xten linker sequence (green). The arrows indicate predicted protease cleavage sites.

(B) Single-well validation of low-activity β-MSH mutants in MC4R HTLA reporter cells. Two nucleotide variants are tested for each β-MSH mutant, and every experiment is done in triplicate. All luciferase measurements are normalized to the average luciferase level of the empty vector control.

(C) Dose-response curve of luciferase activity from MC4R HTLA cells treated with wild-type β-MSH and shortened wild-type β-MSH. MC4R HTLA cells treated with a random 22-mer peptide are used as a negative control. All values are normalized to DMSO-treated cells. Note that all peptide concentrations are expressed on a logarithmic scale by applying the transformation $\log_{10}(\text{peptide concentration} + 0.0001)$.

(D) Single-well validation of high-activity β-MSH mutants in MC4R HTLA reporter cells. Two nucleotide variants are tested for each β-MSH mutant, and every experiment is done in triplicate. All luciferase measurements are normalized to the average luciferase level of the empty vector control.

(E) Dose-response curve of luciferase activity from MC4R HTLA cells treated with wild-type β-MSH and D5H β-MSH mutant. MC4R HTLA cells treated with random 22-mer peptide are used as a negative control. All values are normalized to DMSO-treated cells. Note that all peptide concentrations are expressed on a logarithmic scale by applying the transformation $\log_{10}(\text{peptide concentration} + 0.0001)$.

negatively charged amino acids, respectively. It is possible that the substitution of aspartic acid [D] with histidine [H], which switches a negative charge to a positive charge, facilitates the formation of a stabilizing ionic interaction. This possible stabilizing interaction may account for the observed improvement in the activity of the peptide (Figure S3C). Taken together, the results of this analysis highlight that structural predictions, which are informative, might not be able to completely explain sequence-function relationships. This further underscores the importance of functional screening systems like the peptide display system.

DISCUSSION

A better understanding of peptide-GPCR interactions is important in advancing GPCR biology and enabling drug discovery. In this study, we developed a novel high-throughput peptide display platform that directly links the functional activation of GPCRs to specific peptide variants. Using this system, we conducted a deep mutational screen of the β -MSH peptide, and from a pool of 2,500 peptide variants, we identified several that significantly impact the activation of the MC4R, a key regulator of energy homeostasis and appetite control.⁴ Notably, we discovered a novel D5H mutation of β -MSH that exhibits an increased ability to activate MC4R. Our work aligns with previous findings that certain point mutations in the β -MSH peptide can influence MC4R activation²⁹ and highlights the potential of our platform to identify peptide alterations that can modulate receptor activity more effectively. By demonstrating enhanced MC4R activation through the D5H β -MSH variant, our study suggests potential therapeutic avenues for conditions like obesity.

Compared with previous methods for studying GPCR-peptides, many of which have been critical in furthering our understanding of GPCR biology, the functional readout of our reporter system, combined with the peptide library, enables higher-throughput screening while minimizing the workload for hit identification.^{17,19,22,23} The use of a β -arrestin-based reporter with the peptide display system provides a direct and specific readout of GPCR activation, allowing us to accurately link peptide binding to functional receptor responses within a cell. Integration with oligonucleotide library synthesis enables the rapid generation of a vast library of peptide variants, facilitating comprehensive screening that was previously difficult in cell-based assays. Our platform allows parallel testing of numerous peptides, significantly increasing throughput while reducing time and resource requirements.

Our study presents a novel approach to studying GPCR-peptide interactions, overcoming significant limitations of previous methods. The identification of peptide variants like the D5H β -MSH mutation that enhances MC4R activation underscores the potential of our platform in advancing GPCR research and drug discovery. These findings contribute to a deeper understanding of energy homeostasis mechanisms and offer promising directions for therapeutic interventions in metabolic disorders.

Limitations of this study

It is important to consider the technical limitations of the peptide display platform. Variations in peptide expression levels, influ-

enced by codon usage, can affect the activation of the reporter system (Figure 2D; Data S2). We hypothesize that this may explain why some previously reported putative loss-of-function SNPs did not appear in our screen.²⁹ One way we addressed this limitation in our screen is to include multiple codon variants of the same peptide in our library. While the cell-based reporter system is effective for screening, further *in vivo* assessment of candidate peptides is crucial to understanding the physiological relevance of mutant β -MSH peptides. Although our preliminary testing has demonstrated the applicability of the peptide display system to a few other GPCRs, additional experimental validation is needed to fully assess the broader applicability of this platform to other GPCRs and GPCR variants (Figure S1). Lastly, the tethered peptide's limited degrees of freedom due to the C-terminal membrane presents a challenge, particularly for peptides with GPCR-binding epitopes at the C terminus, which may be less likely to appear as hits.

RESOURCE AVAILABILITY

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Michael C. Bassik (bassik@stanford.edu).

Materials availability

Plasmids generated as part of this study for measuring MC4R activity (pAL63) and cloning libraries of peptides for surface display (pAL57) will be available via Addgene.

Data and code availability

- All sequencing output files are included in Data S2.
- Alignment and screen analysis was performed with previously described HT-recruit pipeline: www.github.com/bintulab/HT-recruit-Analyze.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

A.L., A.S.B., and M.C.B. conceptualized and designed the methodology for the project. A.S.B. and M.C.B. acquired funding for the project, were responsible for project administration, and supervised the project. A.L., K.S., A.N., and N.E. designed and performed the experiments. A.L., J.W., and R.V. performed the computational analyses. A.L., A.S.B., and M.C.B. wrote the manuscript.

DECLARATION OF INTERESTS

Stanford University has filed a patent based on the findings in this manuscript. A.S.B. is on the scientific advisory boards of Cantata Bio and Caribou Biosciences and is on the scientific advisory board of and is an observer on the board of directors of Stylus Medicine. M.C.B. declares outside interest in DEM Biopharma and Stylus Medicine.

STAR★METHODS

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SUPPLEMENTAL INFORMATION

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REFERENCES

1. Yang, D., Zhou, Q., Labroska, V., Qin, S., Darbalaei, S., Wu, Y., Yuliantie, E., Xie, L., Tao, H., Cheng, J., et al. (2021). G protein-coupled receptors: structure- and function-based drug discovery. *Signal Transduct. Target. Ther.* 6, 7.
2. Hauser, A.S., Attwood, M.M., Rask-Andersen, M., Schiöth, H.B., and Gioram, D.E. (2017). Trends in GPCR drug discovery: new agents, targets and indications. *Nat. Rev. Drug Discov.* 16, 829–842.
3. Davenport, A.P., Scully, C.C.G., de Graaf, C., Brown, A.J.H., and Maguire, J.J. (2020). Advances in therapeutic peptides targeting G protein-coupled receptors. *Nat. Rev. Drug Discov.* 19, 389–413.
4. Tao, Y.-X. (2010). The melanocortin-4 receptor: physiology, pharmacology, and pathophysiology. *Endocr. Rev.* 31, 506–543.
5. Wikberg, J.E.S., and Mutulis, F. (2008). Targeting melanocortin receptors: an approach to treat weight disorders and sexual dysfunction. *Nat. Rev. Drug Discov.* 7, 307–323.
6. Laiho, L., and Murray, J.F. (2022). The multifaceted melanocortin receptors. *Endocrinology* 163, bqac083.
7. Tallam, L.S., da Silva, A.A., and Hall, J.E. (2006). Melanocortin-4 receptor mediates chronic cardiovascular and metabolic actions of leptin. *Hypertension* 48, 58–64.
8. Clément, K., van den Akker, E., Argente, J., Bahm, A., Chung, W.K., Connors, H., De Waele, K., Farooqi, I.S., Gonneau-Lejeune, J., Gordon, G., et al. (2020). Efficacy and safety of setmelanotide, an MC4R agonist, in individuals with severe obesity due to LEPR or POMC deficiency: single-arm, open-label, multicentre, phase 3 trials. *Lancet Diabetes Endocrinol.* 8, 960–970.
9. Collet, T.-H., Dubern, B., Mokrosinski, J., Connors, H., Keogh, J.M., Mendes de Oliveira, E., Henning, E., Poitou-Bernert, C., Oppert, J.-M., Tounian, P., et al. (2017). Evaluation of a melanocortin-4 receptor (MC4R) agonist (Setmelanotide) in MC4R deficiency. *Mol. Metab.* 6, 1321–1329.
10. Lotta, L.A., Mokrosiński, J., Mendes de Oliveira, E., Li, C., Sharp, S.J., Luan, J., Brouwers, B., Ayinampudi, V., Bowker, N., Kerrison, N., et al. (2019). Human gain-of-function MC4R variants show signaling bias and protect against obesity. *Cell* 177, 597–607.e9.
11. Metzger, P.J., Zhang, A., Carlson, B.A., Sun, H., Cui, Z., Li, Y., Jahnke, M.T., Layton, D.R., Gupta, M.B., Liu, N., et al. (2024). A human obesity-associated MC4R mutation with defective Gq/11 α signaling leads to hyperphagia in mice. *J. Clin. Investig.* 134, e165418. <https://doi.org/10.1172/JCI165418>.
12. Zeng, R., Zhang, Y., and Du, P. (2014). SNPs of melanocortin 4 receptor (MC4R) associated with body weight in Beagle dogs. *Exp. Anim.* 63, 73–78.
13. Farooqi, I.S., Keogh, J.M., Yeo, G.S.H., Lank, E.J., Cheetham, T., and O'Rahilly, S. (2003). Clinical spectrum of obesity and mutations in the melanocortin 4 receptor gene. *N. Engl. J. Med.* 348, 1085–1095.
14. Aykut, A., Özen, S., Gökşen, D., Ata, A., Onay, H., Atik, T., Darcan, Ş., and Özkinay, F. (2020). Melanocortin 4 receptor (MC4R) gene variants in children and adolescents having familial early-onset obesity: genetic and clinical characteristics. *Eur. J. Pediatr.* 179, 1445–1452.
15. Trier, C., Hollensted, M., Schnurr, T.M., Lund, M.A.V., Nielsen, T.R.H., Rui, G., Andersson, E.A., Svendsstrup, M., Bille, D.S., Gjesing, A.P., et al. (2021). Obesity treatment effect in Danish children and adolescents carrying Melanocortin-4 Receptor mutations. *Int. J. Obes.* 45, 66–76.
16. Cwirla, S.E., Balasubramanian, P., Duffin, D.J., Wagstrom, C.R., Gates, C.M., Singer, S.C., Davis, A.M., Tansik, R.L., Mattheakis, L.C., Boytos, C.M., et al. (1997). Peptide agonist of the thrombopoietin receptor as potent as the natural cytokine. *Science* 276, 1696–1699.
17. Wrighton, N.C., Farrell, F.X., Chang, R., Kashyap, A.K., Barbone, F.P., Mulcahy, L.S., Johnson, D.L., Barrett, R.W., Jolliffe, L.K., and Dower, W. J. (1996). Small peptides as potent mimetics of the protein hormone erythropoietin. *Science* 273, 458–464.
18. Omidfar, K., and Daneshpour, M. (2015). Advances in phage display technology for drug discovery. *Expert Opin. Drug Discov.* 10, 651–669.
19. Molek, P., Strukelj, B., and Bratkovic, T. (2011). Peptide phage display as a tool for drug discovery: targeting membrane receptors. *Molecules* 16, 857–887.
20. Jaroszewicz, W., Morcinek-Orłowska, J., Pierzynowska, K., Gaffke, L., and Węgrzyn, G. (2022). Phage display and other peptide display technologies. *FEMS Microbiol. Rev.* 46, fuab052.
21. Mullen, L.M., Nair, S.P., Ward, J.M., Rycroft, A.N., and Henderson, B. (2006). Phage display in the study of infectious diseases. *Trends Microbiol.* 14, 141–147.
22. Kroeze, W.K., Sassano, M.F., Huang, X.-P., Lansu, K., McCorvy, J.D., Giguère, P.M., Sciaky, N., and Roth, B.L. (2015). PRESTO-Tango as an open-source resource for interrogation of the druggable human GPCRome. *Nat. Struct. Mol. Biol.* 22, 362–369.
23. Foster, S.R., Hauser, A.S., Vedel, L., Strachan, R.T., Huang, X.-P., Gavin, A.C., Shah, S.D., Nayak, A.P., Haugaard-Kedström, L.M., Penn, R.B., et al. (2019). Discovery of human signaling systems: Pairing peptides to G protein-coupled receptors. *Cell* 179, 895–908.e21.
24. Huang, Y., Wiedmann, M.M., and Suga, H. (2019). RNA display methods for the discovery of bioactive macrocycles. *Chem. Rev.* 119, 10360–10391.

25. Merz, M.L., Habeshian, S., Li, B., David, J.-A.G.L., Nielsen, A.L., Ji, X., Il Khwildy, K., Duany Benitez, M.M., Phothirath, P., and Heinis, C. (2024). De novo development of small cyclic peptides that are orally bioavailable. *Nat. Chem. Biol.* **20**, 624–633.
26. Rojas-Chaverra, N.M., Imamura, R., Sato, H., Passioura, T., Mihara, E., Nishimura, T., Takagi, J., Suga, H., Matsumoto, K., and Sakai, K. (2024). A cyclic peptide-grafted Fc with hepatocyte growth factor functionality ameliorates hepatic fibrosis in a non-alcoholic steatohepatitis mouse model. *iScience* **27**, 110426.
27. Lubrano-Berthelie, C., Cavazos, M., Dubern, B., Shapiro, A., Stunff, C.L.E., Zhang, S., Picart, F., Govaerts, C., Froguel, P., Bougneres, P., et al. (2003). Molecular genetics of human obesity-associated MC4R mutations. *Ann. N. Y. Acad. Sci.* **994**, 49–57.
28. Loos, R.J.F., Lindgren, C.M., Li, S., Wheeler, E., Zhao, J.H., Prokopenko, I., Inouye, M., Freathy, R.M., Attwood, A.P., Beckmann, J.S., et al. (2008). Common variants near MC4R are associated with fat mass, weight and risk of obesity. *Nat. Genet.* **40**, 768–775.
29. Lee, Y.S., Challis, B.G., Thompson, D.A., Yeo, G.S.H., Keogh, J.M., Madonna, M.E., Wraight, V., Sims, M., Vatin, V., Meyre, D., et al. (2006). A POMC variant implicates beta-melanocyte-stimulating hormone in the control of human energy balance. *Cell Metab.* **3**, 135–140.
30. Liu, S., Liu, S., Zhu, X., Liang, H., Cao, A., Chang, Z., and Lai, L. (2007). Nonnatural protein-protein interaction-pair design by key residues grafting. *Proc. Natl. Acad. Sci. USA* **104**, 5330–5335.
31. Wade, K.H., Lam, B.Y.H., Melvin, A., Pan, W., Corbin, L.J., Hughes, D.A., Rainbow, K., Chen, J.-H., Duckett, K., Liu, X., et al. (2021). Loss-of-function mutations in the melanocortin 4 receptor in a UK birth cohort. *Nat. Med.* **27**, 1088–1096.
32. Barnea, G., Strapps, W., Herrada, G., Berman, Y., Ong, J., Kloss, B., Axel, R., and Lee, K.J. (2008). The genetic design of signaling cascades to record receptor activation. *Proc. Natl. Acad. Sci. USA* **105**, 64–69.
33. Yueh, Y.G., Yaworsky, P.J., and Kappen, C. (2000). Herpes simplex virus transcriptional activator VP16 is detrimental to preimplantation development in mice. *Mol. Reprod. Dev.* **55**, 37–46.
34. Urlinger, S., Baron, U., Thellmann, M., Hasan, M.T., Bujard, H., and Hillen, W. (2000). Exploring the sequence space for tetracycline-dependent transcriptional activators: novel mutations yield expanded range and sensitivity. *Proc. Natl. Acad. Sci. USA* **97**, 7963–7968.
35. Roney, I.J., Rudner, A.D., Couture, J.-F., and Kærn, M. (2016). Improvement of the reverse tetracycline transactivator by single amino acid substitutions that reduce leaky target gene expression to undetectable levels. *Sci. Rep.* **6**, 27697.
36. Tycko, J., Van, M.V., Aradhana, DelRosso, N., Ye, H., Yao, D., Valbuena, R., Vaughan-Jackson, A., Xu, X., Ludwig, C., et al. (2024). Development of compact transcriptional effectors using high-throughput measurements in diverse contexts. *Nat. Biotechnol.* Published online November 1, 2024. <https://doi.org/10.1038/s41587-024-02442-6>.
37. Biebertmann, H., Castañeda, T.R., van Landeghem, F., von Deimling, A., Escher, F., Brabant, G., Hebebrand, J., Hinney, A., Tschöp, M.H., Grüters, A., and Krude, H. (2006). A role for β -melanocyte-stimulating hormone in human body-weight regulation. *Cell Metab.* **3**, 141–146.
38. Chen, X., Zaro, J.L., and Shen, W.-C. (2013). Fusion protein linkers: property, design and functionality. *Adv. Drug Deliv. Rev.* **65**, 1357–1369.
39. Seth Chhabra, E., Moore, N., Furcht, C., Holthaus, A.M., Liu, J., Liu, T., Schellenberger, V., Kulman, J., Salas, J., and Peters, R. (2015). Evaluation of enhanced in vitro plasma stability of a novel long acting recombinant FVIII-Fc-VWF-XTEN fusion protein. *Blood* **126**, 2279.
40. Guillinger, J.P., Thompson, D.B., and Liu, D.R. (2014). Fusion of catalytically inactive Cas9 to FokI nuclease improves the specificity of genome modification. *Nat. Biotechnol.* **32**, 577–582.
41. Tycko, J., DelRosso, N., Hess, G.T., Aradhana, Banerjee, A., Mukund, A., Van, M.V., Ego, B.K., Yao, D., Spees, K., et al. (2020). High-throughput discovery and characterization of human transcriptional effectors. *Cell* **183**, 2020–2035.e16.
42. Zulkower, V., and Rosser, S. (2020). DNA Chisel, a versatile sequence optimizer. *Bioinformatics* **36**, 4508–4509.
43. Challis, B.G., Pritchard, L.E., Creemers, J.W.M., Delplanque, J., Keogh, J.M., Luan, J., Wareham, N.J., Yeo, G.S.H., Bhattacharyya, S., Froguel, P., et al. (2002). A missense mutation disrupting a dibasic prohormone processing site in pro-opiomelanocortin (POMC) increases susceptibility to early-onset obesity through a novel molecular mechanism. *Hum. Mol. Genet.* **11**, 1997–2004.
44. Harno, E., Galí Ramamoorthy, T., Coll, A.P., and White, A. (2018). POMC: The physiological power of hormone processing. *Physiol. Rev.* **98**, 2381–2430.
45. Teufel, F., Almagro Armenteros, J.J., Johansen, A.R., Gislason, M.H., Pihl, S.I., Tsirigos, K.D., Winther, O., Brunak, S., von Heijne, G., and Nielsen, H. (2022). SignalP 6.0 predicts all five types of signal peptides using protein language models. *Nat. Biotechnol.* **40**, 1023–1025.
46. Abramson, J., Adler, J., Dunger, J., Evans, R., Green, T., Pritzel, A., Ronneberger, O., Willmore, L., Ballard, A.J., Bambrick, J., et al. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493–500.
47. Bertagna, X., Lenne, F., Comar, D., Massias, J.F., Wajcman, H., Baudin, V., Luton, J.P., and Girard, F. (1986). Human beta-melanocyte-stimulating hormone revisited. *Proc. Natl. Acad. Sci. USA* **83**, 9719–9723.
48. Heyder, N.A., Kleinau, G., Speck, D., Schmidt, A., Paisdzior, S., Szczepek, M., Bauer, B., Koch, A., Gallandi, M., Kwiatkowski, D., et al. (2021). Structures of active melanocortin-4 receptor-Gs-protein complexes with NDP- α -MSH and setmelanotide. *Cell Res.* **31**, 1176–1189.
49. Liu, Z., and Hruby, V.J. (2022). MC4R biased signalling and the conformational basis of biological function selections. *J. Cell Mol. Med.* **26**, 4125–4136.
50. Howard, C.J., Abell, N.S., Osuna, B.A., Jones, E.M., Chan, L.Y., Chan, H., Artis, D.R., Asfaha, J.B., Bloom, J.S., Cooper, A.R., et al. (2025). High-resolution deep mutational scanning of the melanocortin-4 receptor enables target characterization for drug discovery. *eLife* **13**, RP104725.
51. Longo, P.A., Kavan, J.M., Kim, M.-S., and Leahy, D.J. (2013). Transient mammalian cell transfection with polyethylenimine (PEI). *Methods Enzymol.* **529**, 227–240.
52. Schubert, M., Lindgreen, S., and Orlando, L. (2016). AdapterRemoval v2: rapid adapter trimming, identification, and read merging. *BMC Res. Notes* **9**, 88.
53. Zhang, J.H., Chung, T.D., and Oldenburg, K.R. (1999). A simple statistical parameter for use in evaluation and validation of high throughput screening assays. *J. Biomol. Screen* **4**, 67–73.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Anti-firefly luciferase antibody	AbCam	cat no. ab21176; RRID: AB_446076
Anti-rabbit conjugated FITC antibody	ThermoFisher Scientific	cat no. 31583; RRID: AB_2787210
Anti-Myc Alexa 488 antibody	AbCam	cat no. ab202008; RRID: AB_3105899
Human TruStain FcX	BioLegend	cat no. 422301
Bacterial and virus strains		
One Shot Stbl3	ThermoFisher Scientific	cat no. C737303
Chemicals, peptides, and recombinant proteins		
Blasticidin S HCl	ThermoFisher Scientific	cat no. A113903
β-MSH peptide	Genscript	N/A
motilin peptide	Genscript	cat no. RP10527
neuropeptide Y	Genscript	cat. no. RP10492
RFRP-1	MedChem Express	cat. no. HY-P1428
Critical commercial assays		
ONE-Glo™ Luciferase Assay	Promega	cat no. E6110
BD Cytotfix/Cytoperm™	BD Bioscience	cat no. 554714
Deposited data		
β-MSH DMS screen	This paper; Data S2	N/A
Experimental models: Cell lines		
HTLA cells	Barnea et al. ³²	N/A
HTLA MC4R-tTA Advance-NFZ	This paper	N/A
HEK293T Lenti-X cells	Takara	632180
Oligonucleotides		
β-MSH peptide library	Twist; see Table S2	N/A
Primers	IDT; see Tables S1 and S3	N/A
Recombinant DNA		
pAL63 MC4R-tTA Advance-NFZ	This paper	Addgene #228457
pAL57_mcherry_hygro_peptide_library	This paper	Addgene #228456
pMD2.G	Didier Trono	Addgene #12259
pRSV-Rev	Didier Trono	Addgene #12253
pMDLg/pRRE	Didier Trono	Addgene #12251
Software and algorithms		
GraphPad Prism	N/A	v10.2.2
CellEngine Software	cellengine.com	N/A
HT-recruit Analyze	https://github.com/bintulab/HT-recruit-Analyze	N/A

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Cell line and cell culture

HTLA cells, a HEK293-derived cell line with stable integrations of a tTA-dependent luciferase reporter and a β-arrestin2-TEV fusion gene, were kindly provided by Dr. Jonathan Long's Laboratory at Stanford University.³² These cells were not authenticated. All cells were cultured in a humidity-controlled incubator at 37°C and 5% CO₂ in DMEM supplemented with 10% fetal bovine serum (FBS) (Sigma-Aldrich, cat no. F6765-500ML), 2 mM L-glutamine (Gibco, cat no. 25030081), and 100 U/ml Penicillin-Streptomycin (Gibco, cat no. 15-140-163).

Plasmids

All primers and gene fragments used in this study were purchased from Integrated DNA Technologies (IDT, [Table S3](#)). Plasmids were constructed using Golden Gate Assembly (New England Biolabs, cat no. E1602L) and transformed into One Shot Stbl3 chemically competent *e. coli* cells (ThermoFisher Scientific, cat no. C737303).

The reporter plasmid for the MC4R-tTA Advance-NFZ gene was derived from the MC4R vector in the Presto-Tango system (Addgene, Kit #1000000068). Briefly, a tetracycline-controlled transactivator with improved stability and reduced noise (tTA Advance) was fused to the MC4R gene.³⁴ To reduce cellular toxicity and increase the measurable signal of MC4R activation, a compact human activator combination NFZ was used in the construct instead of VP16.³⁶ The resulting fusion gene was cloned into a lentiviral vector under an EF-1 α promoter with a blasticidin resistance gene.

For the construction of the plasmid carrying the peptide display vector, DNA fragments encoding a preprotrypsin-derived secretory signal peptide sequence (MSALLILALVGAAVA), a peptide of interest, a myc-tag, a linker sequence, and the transmembrane domain of the platelet-derived growth factor receptor β were assembled and cloned into a lentiviral vector with mCherry and hygromycin resistance genes. The peptide display gene is under an EF-1 α promoter. To simplify the protocol for cloning peptide libraries, the peptide display vector was later modified to include two typellS BsmBI enzyme recognition sites between the secretory signal and the Myc tag.

METHOD DETAILS

Plasmid transfection and transduction

HEK293T Lenti-X cells (Takara, cat no. 632180) were transfected using the polyethyleneimine (PEI) (ThermoFisher Scientific, cat no. NC1014320).⁵¹ Viral supernatants were collected 48 and 72 h post-transfection, filtered through a 0.45 μ m syringe, and applied directly to cells with 4 μ g/mL polybrene (ThermoFisher Scientific, cat no. TR1003G) or stored at -80°C for later use.

Stable MC4R reporter cell line generation

100,000 HTLA cells were plated in a well of a 6-well plate, allowed to attach overnight, and then transduced with lentiviral particles carrying expression vectors for the MC4R-tTA Advance-NFZ (pAL63, available via Addgene) fusion gene. After a media change and 3 days of recovery, the cells were moved into a 10-cm plate and selected with 10 μ g/mL of blasticidin S HCl (ThermoFisher Scientific, cat no. A113903). In parallel, untransduced HTLA cells were treated with equal concentrations of blasticidin to determine when selection was complete. Post selection, the expression of MC4R-TetR-NFZ was validated by treating the HTLA MC4R reporter cells with 500 nM of synthetic β -MSH peptide (Genscript), the putative agonist of the G-protein coupled receptor. MC4R activation was measured using the ONE-Glo Luciferase Assay (Promega, cat no. E6110).

Peptide display assay

To measure MC4R activation by the cis-displayed peptides, 100,000 MC4R HTLA reporter cells were plated in each well of a 6-well plate and allowed to attach overnight. In parallel, lentivirus encoding peptides of interest on the display vector were titrated on HEK293T cells and volumes equating to equal multiplicity of infection (MOI) were calculated. After attaching, MC4R HTLA cells were transduced with the peptide display constructs at equal MOI. Media was changed the next day and cells were given 48 h to recover. MC4R activation was measured using the ONE-Glo Luciferase Assay (Promega, cat no. E6110). As positive controls, the same peptide sequences were synthesized and reconstituted in DMSO. Cells were then treated with 500 nM or 1 μ M of these synthetic peptides [β -MSH peptide (Genscript), motilin peptide (Genscript, RP10527–0.5), neuropeptide Y (Genscript, RP10492–0.5), RFRP-1 (MedChem Express, HY-P1428)].

Cell surface staining to measure peptide expression

Surface staining against the Myc tag was performed to confirm the membrane localization of the peptide display construct. HTLA cells were transfected or transduced with the peptide display vector at least 48 h before staining and proper mCherry expression was confirmed. To measure peptide expression, one million cells were incubated in 100 μ L of staining buffer with 5 μ L of Human TruStain FcX (BioLegend, cat no. 422301) for 5–10 min at room temperature to block the FC-receptor. Cells were probed using Alexa 488 conjugated anti-Myc tag antibody (1 μ L per sample, AbCam, cat no. ab202008). Surface protein expression was measured using flow cytometry (Attune NxT) and gated on mCherry-positive cells. The data was analyzed using the Cell Engine software.

Intracellular staining of luciferase protein

To prepare the cells for intracellular staining, the following steps were taken: Cells were trypsinized, pelleted at 350g at 4°C , and washed in staining buffer (BioLegend, cat no. 420201). Cells were then fixed and permeabilized with 200 μ L of FixPerm solution (BD Bioscience, cat no. 554714). After a 30-min incubation at room temperature, the cells were centrifuged and washed twice with 500 μ L of Permash buffer. Concurrently, the anti-firefly luciferase antibody (abcam cat no. ab21176) was diluted 1:200 in Permash buffer (BD Bioscience, cat no. 554714) and added to each sample following the second wash. After a 1-h incubation, excess antibodies were removed with further washes in Permash buffer. Cells were then incubated with a secondary anti-rabbit conjugated FITC antibody at a 1:500 dilution for another hour (ThermoFisher Scientific, cat no. 31583). Luciferase expression was quantified using flow cytometry (Attune NxT), focusing on mCherry-positive cells, and data was analyzed using Cell Engine software. This protocol was designed for up to 5 million cells and was scaled proportionally to accommodate larger cell numbers in the library screen.

Design of β -MSH mutagenesis library

An all-by-all deep mutational scan of the 22 amino acid β -MSH peptide sequence (AEKKDEGPYRMEHFRWGSPPKD) was performed to generate 418 mutant peptides. Each peptide was then encoded 5 times with varied codon usage, for a total of 2090 mutant elements. Primer binding and tyellIS BsmBI enzyme recognition sites were concatenated to the 5' and 3' ends of each mutant element. Random DNA padding sequences were generated and added between the 5' primer binding and 5' enzyme recognition sites to increase oligo size to 260 bp. 30 wild type encodings and 30 random encodings were created as positive and negative controls, respectively. The final library consisted of 2150 elements, each 260 bp in length. The first 120 bp of each oligo was required to be unique to ensure we would be able to distinguish every library element with short-read sequencing. Additionally, 7xC homopolymers and internal BsmBI sites were removed, 20-75% GC content was required in every 50 bp window, and human codon usage was optimized in every oligo using DNAAchisel.⁴²

Peptide library cloning

Oligonucleotides encoding the peptides were synthesized in the form of a pooled library through Twist Biosciences. PCR amplification was conducted in a controlled PCR hood environment to prevent unintentional amplification of contaminating DNA. For each of the six reactions, we used 5 ng of template, 0.1 mL of each 100 mM primer, 1 mL of Herculase II polymerase (from Agilent), 1 mL of DMSO, 1 mL of 10 nM dNTPs, and 10 mL of 5x Herculase buffer. The thermocycling protocol consisted of an initial step at 98°C for 3 min, followed by a series of cycles at 98°C for 20 s, 61°C for 20 s, 72°C for 30 s, and a final extension step at 72°C for 3 min. The default cycle number was set at 29, with adjustments made to determine the minimum number of cycles required to obtain a clean and visible product for subsequent gel extraction (typically, 25 cycles proved to be the minimum necessary). Following PCR, the resulting double-stranded DNA libraries underwent gel extraction, where the desired bands at approximately 250 base pairs were excised from a 2% TBE gel (four lanes), and a QIAgen gel extraction kit was employed for purification. Due to the nature of the varying length of the peptide encoding oligos, the PCR product was not a clear single band but a smear ranging from 400 to 200 base pairs and therefore, the whole smear was extracted. These libraries were subsequently cloned into the lentiviral peptide display vector (pAL57, available via Addgene) using a total of 4 × 10 mL GoldenGate reactions, which included 75 ng of pre-digested and gel-extracted backbone plasmid, 5 ng of library (in a 2:1 molar ratio of insert to backbone), 0.13 mL of T4 DNA ligase (NEB, 20000 U/ml), 0.75 mL of Esp3I-HF (NEB), and 1 mL of 10x T4 DNA ligase buffer. These reactions underwent 30 cycles of digestion at 37°C and ligation at 16°C for 5 min each, followed by a final 5-min digestion at 37°C and heat inactivation at 70°C for 20 min. The resulting reactions were pooled and purified using MinElute columns (QIAgen), with an elution volume of 6 μ L of ddH₂O. Subsequently, 2 mL per tube was transformed into two tubes containing 50 mL of Endura electrocompetent cells (Lucigen, cat no. #60242-2) following the manufacturer's instructions. After the recovery process, the cells were plated on 3 - 7 large 10" x 10" LB plates with carbenicillin. Following overnight growth at 37°C, the bacterial colonies were scraped into a collection bottle, and plasmid pools were extracted using a HiSpeed Plasmid Maxiprep kit (QIAgen). In parallel, 2 - 3 small plates were prepared with diluted transformed cells to count colonies and confirm the transformation efficiency, ensuring at least 30x library coverage was maintained. To assess the quality of the libraries, 20 - 30 colonies from the transformations underwent Sanger sequencing (Quintara) and the peptides were amplified from the plasmid pool for premium PCR (Primordium) to estimate cloning efficiency and the proportion of empty backbone plasmids within the pools. The PCR and sequencing protocols were consistent with those described below for sequencing from genomic DNA. The resulting sequencing datasets were analyzed, as detailed below, to assess coverage uniformity and synthesis quality of the libraries.

High throughput β -MSH peptide mutagenesis screen

Large-scale lentivirus production and infection of MC4R HTLA cells were performed as follows: To generate sufficient lentivirus to infect the libraries into MC4R HTLA cells, HEK293T cells were plated on two 15-cm tissue culture plates. On each plate, 9 million HEK293T cells were plated in 30 mL of DMEM, grown overnight, and then transfected with 8 mg of an equimolar mixture of the three third-generation packaging plasmids (pMD2.G, psPAX2, pMDLg/pRRE) and 8 mg of β -MSH peptide library vectors using 50 mL of polyethylenimine (PEI, Polysciences #23966). pMD2.G (Addgene, plasmid #12259), psPAX2 (Addgene, plasmid #12260), and pMDLg/pRRE (Addgene plasmid #12251). After 48 h and 72 h of incubation, lentivirus was harvested and filtered through a 0.45-mm PVDF filter (Millipore) to remove any cellular debris. For the β -MSH peptide screen, approximately 5 million MC4R HTLA cells were seeded into each 15-cm tissue culture plate. We prepared four plates, constituting two biological replicates. The cells were allowed to attach overnight before being infected with lentiviruses encoding the peptide library at an MOI of approximately 0.3. This infection rate was chosen to ensure a higher-than-standard coverage of over 1000 for each library element, to offset potential losses during subsequent FACS sorting, library preparation, and synthesis errors. After 3 days of recovery, the infected cells were trypsinized, and mCherry measurements were taken to confirm the efficiency of the infection. The cells were then replated at a density of 5 million cells per 15-cm plate and allowed to grow for an additional 2 days. Subsequently, the cells were split and stained for luciferase protein expression in preparation for fluorescence-activated cell sorting (FACS). FACS was conducted at the Stanford FACS facility, where cells were initially gated on mCherry to isolate those expressing the peptide library. Further gating was done based on luciferase expression, as indicated by the intensity of the FITC signal. Cells falling within the top or bottom 15% based on FITC signal intensity were sorted into separate bins. Cells were pelleted and frozen at -20°C until genomic DNA extractions were performed.

Library preparation and sequencing

Genomic DNA was extracted with the QIAgen Blood Maxi Kit following the manufacturer's instructions. DNA was eluted in EB and not AE to avoid subsequent PCR inhibition. A test PCR was performed using 2.5 µg of genomic DNA in a 25 µL (half size) reaction to verify if the PCR conditions would result in a visible band at the expected size for each sample. After determining the optimal PCR condition, 50 µL reactions were prepared in 96 well plates with the number of reactions depending on the amount of genomic DNA available in each experiment. 5 µg of genomic DNA, 0.25 µL of each 100 mM primer, and 25 µL of NEBnext 2x Master Mix (NEB) were used in each reaction. The reactions were set up on ice in a clean PCR environment to prevent contamination. The PCR protocol involved preheating the thermocycler to 98°C, followed by a 3-min initial denaturation at 98°C, and then 27 cycles of denaturation at 98°C for 10 s, annealing at 63°C for 30 s, extension at 72°C for 30 s, and a final extension at 72°C for 2 min. All subsequent procedures were carried out outside the PCR hood. The PCR reactions were combined, and the pooled material was loaded onto a 2% TBE gel, alongside a 100-bp ladder. Gel electrophoresis was conducted until clear ladder separation was observed, and the peptide library band around 200bp was excised. DNA purification was performed using the QIAquick Gel Extraction Kit (QIAgen). DNA was elution into non-stick tubes (Ambion). A follow-up gel analysis confirmed the removal of small fragments. These libraries were then quantified using a Qubit HS kit (ThermoFisher Scientific) and sequencing using Illumina NextSeq using a single-end forward read (150 cycles) and 8-cycle index reads (Novogene).

Peptide screen analysis

The reads were trimmed using AdapterRemoval.⁵² The subsequent steps follow the standard workflow of the HT-recruit analysis (www.github.com/bintulab/HT-recruit-Analyze). First, a Bowtie reference was created using the designed library sequences via the 'makeIndices.py' script, and the reads were aligned allowing for zero mismatches using the 'makeCounts.py' script. The enrichment differences for each peptide in the top and bottom 15% bins were calculated using the 'makeRhos.py' script. Peptides with fewer than five reads in both samples of a particular replicate were excluded from that replicate and set to zero counts. If peptides had fewer than five reads in one sample, those reads were adjusted to five to prevent the inflation of enrichment values due to low sequencing depth. For all peptide screens, peptides with a count of five or fewer in both replicates for a given condition were removed from downstream analysis. For each peptide, the enrichment difference between the top and bottom 15% bins was log2-transformed. High (and low) activity mutants were defined as having an enrichment score higher (or lower) than two standard deviations above (or lower) the mean for wild type β-MSH peptides. Agreement between replicates were tested using Fisher's exact test for the identification of high and low activity mutants. Additionally, we used the Z prime factor test for signal-to-noise based on the random 22-mers included in the screen.⁵³

Given the high concordance between the two biological replicates (see Figure 2B), values were averaged across the two biological replicates. For residue-based analyses (see Figures 2C and 2D), the enrichment scores for all five codon variants were averaged and Z scores were computed based on the mean and standard deviation for wild type β-MSH peptides.

Peptide activation assay

All the peptides used in this study were synthesized by Genscript. To quantify relative levels of activation of MC4R by a particular peptide, 50,000 MC4R HTLA were seeded in a 24-well plate with 500µL of culturing media. Cells were allowed to attach overnight and fresh media was added to each well. The highest concentration of the peptide was added onto the first column of the plate and then four 10-fold serial dilutions were performed across the plate. After 24 h of treatment, luciferase reporter activation was quantified using the ONE-Glo Luciferase Assay (Promega, cat no. E6110). EC50 values were calculated using a four-parameter agonist vs. concentration model in Prism 10 (Graphpad, San Diego, California). In order to plot the zero value on a logarithmic scale, an offset value of 0.0001 was added to all cell treatment concentrations. Untreated cells were represented by a fixed value of Log₁₀(-4) on the graph.

Peptide-receptor structural analysis

Using AlphaFold 3, we co-folded full length MC4R with the wild type truncated β-MSH peptide and the D5H peptide individually.⁴⁶ Then we downloaded the PDB structure 7PIV that represents the cryo-electron microscopy structure of the NDP-α-MSH–MC4R complex.⁴⁸ Using PyMOL (Version 2.5.2), we aligned the MC4R receptor from the alphafolded structures with the MC4R receptor from the 7PIV PDB entry. Then we generated figures using UCSF ChimeraX (Version 1.9rc202412032254)

QUANTIFICATION AND STATISTICAL ANALYSIS

The screening data was analyzed using Python and R. The workflow and schematic summary of this study were created using BioRender. Statistical significances between groups in luciferase reporter assays were determined using either a two-tailed Student's t test facilitated by GraphPad Prism 10 (GraphPad Software Inc, California, USA). Significance was set at $p < 0.05$. Flow cytometry data was analyzed using the Cell Engine software (www.cellengine.com).